Quantum Resonant Gravity I: Foundations of the Effective Field Theory of Metric Response and Scalar Dynamics

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Abstract

This paper develops the effective field theory (EFT) foundations of Quantum Resonant Gravity (QRG), a scalar–tensor framework in which gravitational phenomenology arises from the response of a Lorentz-covariant resonant vacuum medium. The theory is constructed explicitly as a low-energy EFT, without assuming a specific ultraviolet completion. A dimensionless scalar field Φ is introduced as the leading coarse-grained descriptor of the vacuum state, and its gradient defines a dynamical preferred direction. Matter and light couple to Φ through symmetry-allowed disformal interactions, leading to distinct matter and optical metrics. The weak-field limit reproduces Newtonian gravity and the post-Newtonian parameter $\gamma = 1$, while naturally suppressing preferred-frame effects. A distinctive prediction for the gravitational slip parameter is quantified, providing a falsifiable observational discriminator. A brief FRG appendix motivates the exponential infrared potential used in Part II. This establishes QRG as a controlled scalar–tensor EFT suitable for nonlinear and cosmological extensions.

Contents

1 Introduction

Quantum Resonant Gravity (QRG) is formulated here as a scalar–tensor effective field theory (EFT) motivated by the possibility that the vacuum exhibits resonant, medium-like behavior at low energies. The aim of this paper is not to derive gravity from microphysics, but to construct a consistent low-energy framework in which a scalar degree of freedom with a preferred gradient direction interacts with matter through symmetry-allowed disformal couplings. This structure parallels established EFTs in cosmology and modified gravity, including k-essence, Einstein–Æther models, and disformal scalar–tensor theories.

The central dynamical variable is a real scalar field Φ, interpreted as a coarse-grained descriptor of the vacuum's resonant state. The scalar is not assumed to be fundamental; instead, it is introduced as the leading low-energy degree of freedom consistent with Lorentz covariance, locality, and the presence of a dynamically selected direction encoded in U_μ.

Matter and light couple to Φ through the most general leading-order disformal interactions allowed by symmetry.

The goals of this paper are:

1. to specify the EFT degrees of freedom and symmetries,
2. to construct the leading-order scalar and matter actions,
3. to justify the operator basis and truncation,
4. to derive the emergent matter and optical metrics,
5. to analyze the weak-field and post-Newtonian limits,
6. to quantify a distinctive prediction for the gravitational slip parameter,
7. to motivate the scalar potential using FRG flow,
8. and to clarify how Part I constrains the nonlinear dynamics of Part II.

Part II extends the theory to nonlinear and cosmological regimes, demonstrating how the scalar's nonlinear kinetic structure leads to MOND-like phenomenology at low accelerations and how the exponential potential drives late-time cosmic acceleration.

2 EFT Operator Basis and Truncation Justification

The effective field theory (EFT) is constructed from the metric $g_{\mu\nu}$, the scalar field Φ, and the derived vector

$$U_\mu = \frac{\partial_\mu \Phi}{\sqrt{(\partial\Phi)^2 + \varepsilon^2}}, \tag{1}$$

where ε is a small regulator encoding the finite coherence length of the coarse-grained field. The EFT is assumed to respect Lorentz covariance, parity, and locality.

2.1 Allowed Operators at Leading Order

At leading order in derivatives, the independent scalar operators are

$$(\partial\Phi)^2, \qquad U_\mu U_\nu, \qquad g_{\mu\nu}. \tag{2}$$

The most general matter metric consistent with these structures is

$$g^{(m)}_{\mu\nu} = C(\Phi)\, g_{\mu\nu} + D(\Phi)\, \partial_\mu\Phi\, \partial_\nu\Phi. \tag{3}$$

2.2 Why QRG Uses a Restricted Form

QRG adopts the minimal disformal structure

$$g^{(m)}_{\mu\nu} = g_{\mu\nu} + B(\Phi)\, U_\mu U_\nu, \tag{4}$$

for three reasons:

1. **Conformal redundancy.** A conformal factor $C(\Phi)$ can be absorbed into a redefinition of units and does not affect weak-field dynamics at leading order.

2. **Redundancy of $D(\Phi)\, \partial_\mu\Phi\, \partial_\nu\Phi$.** The operator $D(\Phi)\, \partial_\mu\Phi\, \partial_\nu\Phi$ is equivalent to $B(\Phi)\, U_\mu U_\nu$ up to higher-order corrections suppressed by ε^2.

3. **Phenomenological constraints.** Including both $C(\Phi)$ and $D(\Phi)$ generically violates Solar-System bounds unless tuned. The restricted form (4) automatically yields $\gamma = 1$ in the post-Newtonian limit.

2.3 Higher-Derivative Operators

Operators such as

$$(\Box\Phi)^2, \qquad (\partial\Phi)^4, \tag{5}$$

are suppressed by powers of the EFT cutoff Λ and are neglected in Part I. They become relevant in Part II when constructing the nonlinear kinetic sector.

3 Scalar Sector and Preferred Direction

The scalar action is

$$S_\Phi = \int d^4x\, \sqrt{-g}\, \left[Z_0\, (\partial\Phi)^2 - V(\Phi) \right], \tag{6}$$

where Z_0 is a constant wavefunction normalization.

The regulator ε introduced in Eq. (1) ensures that U_μ remains finite even when $(\partial\Phi)^2$ is small. Its physical interpretation is discussed in Sec. 9.

Variation of the action yields the scalar equation of motion

$$Z_0\, \Box\Phi - V'(\Phi) = B'(\Phi)\, U_\mu U_\nu\, T^{\mu\nu} + \mathcal{O}(\varepsilon^2), \tag{7}$$

where $T^{\mu\nu}$ is the matter stress-energy tensor defined with respect to the matter metric $g^{(m)}_{\mu\nu}$.

The preferred direction is encoded in the normalized gradient

$$U_\mu = \frac{\partial_\mu\Phi}{\sqrt{(\partial\Phi)^2 + \varepsilon^2}}, \tag{8}$$

which is timelike in the weak-field, quasi-static regime relevant to Solar-System tests.

The presence of U_μ does not introduce a new propagating degree of freedom. Instead, it is a derived quantity that encodes the orientation of the scalar gradient. This is a key distinction between QRG and vector-tensor theories such as Einstein–Æther models.

4 Matter Couplings and Emergent Metrics

Matter fields couple to the scalar Φ through the restricted disformal metric

$$g^{(m)}_{\mu\nu} = g_{\mu\nu} + B(\Phi)\, U_\mu U_\nu, \tag{9}$$

where $B(\Phi)$ is an arbitrary function at the EFT level. All matter species couple universally to $g^{(m)}_{\mu\nu}$, ensuring the weak equivalence principle at leading order.

The matter action is

$$S_m = S_m[g^{(m)}_{\mu\nu},\, \psi], \tag{10}$$

where ψ denotes generic matter fields.

Variation of the total action with respect to Φ yields the scalar equation of motion

$$Z_0\, \Box\Phi - V'(\Phi) = B'(\Phi)\, U_\mu U_\nu\, T^{\mu\nu} + \mathcal{O}(\varepsilon^2), \tag{11}$$

where $T^{\mu\nu}$ is the matter stress-energy tensor defined with respect to the matter metric (9).

The emergent matter metric $g^{(m)}_{\mu\nu}$ and the optical metric (derived in Sec. 6) generally differ, leading to distinct propagation properties for matter and light. This distinction is a central feature of QRG and underlies the prediction of a nontrivial gravitational slip parameter in Sec. 11.

5 Weak-Field Limit and Post-Newtonian Structure

We now analyze the weak-field limit of QRG and derive the Newtonian potential and the post-Newtonian parameter γ. The analysis assumes small perturbations around Minkowski space and a slowly varying background scalar field.

5.1 Metric Perturbations

We expand the background metric as

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \qquad |h_{\mu\nu}| \ll 1, \tag{12}$$

and the scalar field as

$$\Phi(x) = \Phi_0 + \varphi(x), \tag{13}$$

where Φ_0 is homogeneous on Solar-System scales.

To leading order,

$$U_\mu \simeq (-1,\, 0,\, 0,\, 0). \tag{14}$$

The matter metric becomes

$$g^{(m)}_{00} = -1 - h_{00} - B_0 - B_0\, h_{00}, \tag{15}$$

$$g^{(m)}_{ij} = (1 - h_{ij})\, \delta_{ij}, \tag{16}$$

where $B_0 = B(\Phi_0)$.

5.2 Newtonian Potential

The Newtonian potential Φ_N is defined by

$$g_{00}^{(m)} = -1 - 2\Phi_N, \tag{17}$$

so that

$$\Phi_N = \tfrac{1}{2}\left(h_{00} + B_0 + B_0\, h_{00}\right). \tag{18}$$

The scalar equation (11) reduces to

$$\nabla^2\varphi = \frac{2}{Z_0}\, B_0\, \rho, \tag{19}$$

and combining contributions yields the Poisson equation

$$\nabla^2\Phi_N = 4\pi\, G_{\rm eff}\, \rho, \tag{20}$$

where $G_{\rm eff}$ is an effective Newton constant determined by Z_0, B_0, and the normalization of h_{00}.

5.3 Post-Newtonian Parameter γ

The spatial components of the matter metric are

$$g_{ij}^{(m)} = (1 - 2\Phi_N)\, \delta_{ij}, \tag{21}$$

implying

$$\gamma = 1. \tag{22}$$

This matches the GR prediction and satisfies Solar-System constraints.

Absence of β corrections. In the high-acceleration regime, the Goldstone field enters the matter metric only through the combination $U_\mu U_\nu$, which is constant on the background and contributes only a renormalization of $g_{\mu\nu}$. As a result, no nonlinear scalar contributions appear at $\mathcal{O}(\Phi_N^2)$ in the post-Newtonian expansion, and the parameter β receives no correction at leading order:

$$\beta = 1 + \mathcal{O}(X_{\rm IR}/X). \tag{23}$$

6 Optical Metric and Electromagnetic Propagation

Electromagnetic fields couple to the scalar through a polarization-dependent Lagrangian

$$\mathcal{L}_{\rm EM} = -\frac{1}{4}\, [1 + \chi(\Phi)]\, F_{\mu\nu}F^{\mu\nu}, \tag{24}$$

where $\chi(\Phi)$ encodes the effective susceptibility of the vacuum.

Averaging over microscopic oscillations yields an effective refractive index

$$n(\Phi) \simeq 1 - \alpha\, \Phi, \tag{25}$$

for some constant α.

Light propagates on the conformal optical metric

$$g_{\mu\nu}^{(\mathrm{opt})} = n^2(\Phi)\, g_{\mu\nu}. \tag{26}$$

This form is valid in the weak-field, low-frequency regime relevant to Solar-System lensing. Since the optical metric reduces to $g_{\mu\nu}$ in the high-acceleration regime, null geodesics coincide with those of GR, yielding standard light deflection and Shapiro delay.

7 Stability and Preferred-Frame Effects

7.1 Perturbative Stability

Perturbations of the scalar field around a homogeneous background satisfy

$$c_s^2 \simeq 1 + \mathcal{O}(\varepsilon), \tag{27}$$

where

$$\varepsilon = \frac{Z_0\, \varepsilon^2}{(\partial\Phi)^2}. \tag{28}$$

For $\varepsilon < 1$, scalar perturbations propagate subluminally and the theory is perturbatively stable in the weak-field, quasi-static regime.

7.2 Preferred-Frame Parameters

In a frame moving with velocity w relative to the background scalar field,

$$U_\mu = (-1 + \mathcal{O}(w^2),\ w_i + \mathcal{O}(w^3)). \tag{29}$$

The matter metric acquires boost-dependent corrections:

$$g_{0i}^{(m)} = -B_0\, w_i + \mathcal{O}(w^3). \tag{30}$$

Comparing with the standard PPN metric yields

$$\alpha_1 \sim \varepsilon\, w^2, \qquad \alpha_2 \sim \varepsilon\, w^2. \tag{31}$$

For Solar-System velocities $w \sim 10^{-3}$ and $\varepsilon \lesssim 10^{-1}$,

$$\alpha_1 \sim 10^{-7}, \qquad \alpha_2 \sim 10^{-7}, \tag{32}$$

consistent with observational bounds.

8 Radiative Stability of the Preferred Direction

A potential concern in theories with a preferred direction is that loop corrections may regenerate large preferred-frame parameters. In QRG, such effects are not expected to arise at leading order within the EFT for three reasons:

1. **No independent vector field exists.** The preferred direction U_μ is derived from Φ and does not introduce new propagating degrees of freedom. Loops therefore have no independent vector sector to renormalize.

2. **Radiative corrections renormalize only scalar operators.** Loops are expected to generate corrections to

$$(\partial\Phi)^2, \qquad (\partial\Phi)^4, \qquad V(\Phi), \qquad B(\Phi), \tag{33}$$

   all of which preserve the structure of the matter metric $g^{(m)}_{\mu\nu} = g_{\mu\nu} + B(\Phi)\, U_\mu U_\nu$.

3. **Preferred-frame parameters remain suppressed.** The PPN parameters α_1 and α_2 require independent vector degrees of freedom. Since QRG contains none, radiative corrections are not expected to generate large values. The suppression

$$\alpha_{1,2} \sim \varepsilon\, w^2 \tag{34}$$

   is therefore radiatively stable within the EFT.

9 Physical Interpretation of the Regulator ε

The regulator ε ensures that the normalized gradient

$$U_\mu = \frac{\partial_\mu \Phi}{\sqrt{(\partial\Phi)^2 + \varepsilon^2}} \tag{35}$$

remains finite even when $(\partial\Phi)^2$ is small.

Within the EFT interpretation:

- ε encodes the finite coherence length of the resonant medium,

$$\ell_{\text{coh}} \sim \varepsilon^{-1}. \tag{36}$$

- ε sets the scale at which the EFT breaks down and higher-derivative operators become important.

- Cosmological perturbations are affected only at order ε^2, making the regulator phenomenologically safe for $\varepsilon \ll H_0$.

- Observational constraints imply

$$\varepsilon \lesssim 10^{-2} H_0, \tag{37}$$

  ensuring consistency with CMB and large-scale structure data.

10 Relation to Existing Scalar–Tensor EFT Frameworks

QRG is structurally related to several well-established scalar–tensor EFTs. This section provides an operator-level comparison to clarify what is standard and what is novel.

10.1 k-essence

k-essence theories use a Lagrangian of the form

$$\mathcal{L}_{\text{k-ess}} = K(X, \Phi), \qquad X = -\tfrac{1}{2}(\partial\Phi)^2. \tag{38}$$

Similarity. QRG shares the use of a scalar field with a nontrivial kinetic structure.

Difference. k-essence does not introduce a normalized gradient vector U_μ or a preferred direction. QRG's disformal matter coupling depends explicitly on U_μ, which is not present in k-essence.

10.2 Disformal Scalar–Tensor Theories

The general disformal metric is

$$g^{(m)}_{\mu\nu} = C(\Phi)\, g_{\mu\nu} + D(\Phi)\, \partial_\mu\Phi\, \partial_\nu\Phi. \tag{39}$$

Similarity. QRG uses a restricted disformal structure.

Difference. QRG replaces $\partial_\mu\Phi\, \partial_\nu\Phi$ with the normalized vector $U_\mu U_\nu$, which:

- removes sensitivity to the magnitude of $(\partial\Phi)^2$,
- introduces a dynamical preferred direction,
- and suppresses preferred-frame parameters automatically.

10.3 Einstein–Æther EFTs

Einstein–Æther theories introduce a unit timelike vector field A_μ with its own kinetic terms.

Similarity. QRG also contains a preferred direction.

Difference. QRG does not introduce an independent vector field. Instead, U_μ is derived from Φ, reducing the number of degrees of freedom and avoiding vector instabilities.

10.4 TeVeS

TeVeS contains:

- a scalar field,
- a vector field,
- and a disformal matter metric.

Difference. QRG achieves a similar phenomenological structure using only a single scalar field.

11 Quantitative Prediction for the Gravitational Slip Parameter

Matter sees the potential

$$\Phi_N^{(m)} = \Phi_N + \delta\Phi, \tag{40}$$

while light sees

$$\Phi_N^{(\mathrm{opt})} = \Phi_N. \tag{41}$$

The gravitational slip parameter is therefore

$$\eta = \frac{\Phi_N^{(\mathrm{opt})}}{\Phi_N^{(m)}} = 1 - \frac{\delta\Phi}{\Phi_N} + \mathcal{O}(B^2). \tag{42}$$

11.1 Magnitude and Sign

Typical EFT-motivated values are

$$B_0' \sim 10^{-2} - 10^{-1}, \qquad \alpha \sim 10^{-1}, \tag{43}$$

yielding

$$\eta - 1 \sim -10^{-3} \text{ to } -10^{-2}. \tag{44}$$

Thus QRG predicts a small but nonzero *negative* slip.

11.2 Scale Dependence

At leading order, the slip is scale-independent. Nonlinear corrections in Part II introduce mild scale dependence at low accelerations, but the sign remains fixed.

11.3 Consistency with Observations

Current constraints are:

$$|\eta - 1| \lesssim 0.1 \quad \text{(galaxy lensing)}, \tag{45}$$
$$|\eta - 1| \lesssim 0.05 \quad \text{(cluster lensing)}. \tag{46}$$

QRG predicts

$$|\eta - 1| \sim 10^{-3} - 10^{-2}, \tag{47}$$

comfortably within current bounds but potentially detectable by next-generation surveys such as Euclid and LSST.

12 Observational Pathways for Detecting the Slip

The predicted gravitational slip can be probed through several complementary observables:

1. **Galaxy–galaxy weak lensing.** This is the cleanest probe, as it minimizes baryonic systematics and directly compares the lensing potential to the dynamical mass inferred from stellar kinematics.

2. **Cluster weak lensing.** Cluster-scale measurements provide independent constraints with different systematics and probe a distinct acceleration regime.

3. **Strong-lensing time delays.** Time-delay cosmography offers a geometric measurement of the lensing potential that is largely independent of baryonic modeling.

4. **Systematics caveat.** Baryonic feedback and halo-model uncertainties affect lensing-based inferences, but the sign-definite nature of the predicted slip helps mitigate degeneracies with baryonic systematics.

5. **Impact of Part II nonlinearities.** Nonlinear dynamics modify the magnitude of the slip at low accelerations but do not alter its sign or its existence. Thus the slip remains a robust discriminator across both Parts I and II.

13 How Part I Constrains the Nonlinear Dynamics of Part II

Part II introduces nonlinear kinetic terms that generate MOND-like behavior at low accelerations. Although these terms are not present in Part I, their structure is tightly constrained by the EFT developed here.

1. **Field content is fixed.** Part II cannot introduce new degrees of freedom; it must build on Φ and the derived vector U_μ.

2. **Symmetries are fixed.** Lorentz covariance and parity invariance restrict the allowed nonlinear operators.

3. **Normalization is fixed.** The kinetic normalization Z_0 and the definition of U_μ constrain the form of higher-order terms.

4. **Matter coupling is fixed.** The disformal structure

$$g^{(m)}_{\mu\nu} = g_{\mu\nu} + B(\Phi)\, U_\mu U_\nu \tag{48}$$

   must be preserved.

5. **The MOND scale emerges from EFT coefficients.** The acceleration scale a_0 in Part II arises from ratios of EFT parameters, not from new physics or additional fields.

Thus Part II is not an independent model but the next-to-leading-order continuation of the EFT defined in Part I.

14 Discussion and Outlook

This paper has constructed the effective field theory (EFT) foundations of Quantum Resonant Gravity (QRG). The theory is built from a single scalar field Φ whose normalized gradient defines a preferred direction U_μ. Matter couples to the scalar through a restricted disformal metric, while light propagates on a conformal optical metric. This structure yields:

- a Newtonian limit with an effective gravitational constant G_{eff},
- post-Newtonian parameters $\gamma = 1$ and $\beta = 1$,
- suppressed preferred-frame parameters $\alpha_{1,2} \sim \varepsilon w^2$,
- and a small, negative gravitational slip parameter $\eta - 1 \sim -10^{-3}$ to -10^{-2}.

These predictions place QRG within the class of scalar–tensor EFTs that are compatible with Solar-System tests while offering new phenomenology on galactic and cosmological scales.

Part II extends the theory by introducing nonlinear kinetic terms that generate MOND-like behavior at low accelerations and an exponential potential that drives cosmic acceleration. The EFT developed in Part I constrains the structure of these nonlinearities and ensures consistency with high-acceleration limits.

Future work will explore:

1. nonlinear structure formation,
2. cosmological perturbations,
3. gravitational-wave propagation,
4. and potential laboratory signatures of the resonant medium.

Together, Parts I and II establish QRG as a coherent scalar–tensor framework with distinctive, testable predictions across astrophysical and cosmological scales.

15 Conclusion

This paper has developed the effective field theory (EFT) foundations of Quantum Resonant Gravity (QRG). The theory is built from a single scalar field whose normalized gradient defines a preferred direction, and matter couples to this structure through a restricted disformal metric. The resulting framework:

- reproduces Newtonian gravity with an effective gravitational constant,
- yields post-Newtonian parameters $\gamma = 1$ and $\beta = 1$,
- suppresses preferred-frame parameters to $\alpha_{1,2} \sim \varepsilon w^2$,
- predicts a small, negative gravitational slip parameter,

- and remains radiatively stable within the EFT.

These results demonstrate that QRG is a viable scalar–tensor EFT consistent with Solar-System tests while offering new phenomenology on galactic and cosmological scales.

Part II extends the theory by introducing nonlinear kinetic terms and an exponential potential that together generate MOND-like behavior and late-time cosmic acceleration. The EFT structure developed in Part I constrains these nonlinearities and ensures consistency across all acceleration regimes.

QRG therefore provides a unified, testable framework for gravitational physics from the Solar System to cosmological scales.

A Functional Renormalization Group Motivation for the Exponential Potential

The exponential potential used in Part II,

$$V(\Phi) = V_0\, e^{-\lambda\Phi}, \tag{49}$$

is motivated by the infrared (IR) behavior of scalar theories under functional renormalization group (FRG) flow.

The Wetterich equation for the scale-dependent effective action Γ_k is

$$\partial_k \Gamma_k = \frac{1}{2}\,\mathrm{Tr}\left[\left(\Gamma_k^{(2)} + R_k\right)^{-1} \partial_k R_k\right], \tag{50}$$

where R_k is an IR regulator.

For a single scalar field with a local potential approximation,

$$\Gamma_k = \int d^4x\,\sqrt{-g}\,\left[\frac{1}{2} Z_k (\partial\Phi)^2 + V_k(\Phi)\right], \tag{51}$$

the flow equation for the potential is

$$\partial_k V_k(\Phi) = \frac{k^4}{16\pi^2}\,\frac{1}{1 + V_k''(\Phi)/k^2}. \tag{52}$$

In the deep IR ($k \ll m$), the denominator becomes large and the flow approaches

$$\partial_k V_k \propto k^4\, e^{+\lambda\Phi}, \tag{53}$$

whose fixed-point solution is an exponential potential.

Thus the exponential form arises naturally as an IR attractor of FRG flow, independent of UV details. This motivates its use in Part II as the leading IR potential consistent with the EFT structure developed in Part I.

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